CN: NFCD6

ENGF0003

Mathematical Modelling and Analysis CO_2 Dissolution in Water

Coursework 2

University College London

ENGF0003

13.12.24

TABLE OF CONTENTS

TASK 1 3

A - Mathematical Task 3

Implications of a Condition Where the Determinant of the Coefficient Matrix Becomes Zero 3

B - Communication Task 4

C – Modelling Task 5

TASK 2 6

A - Coding Task 6

B - Validation Task 6

C - Modelling Task 7

TASK 3 8

A - Coding Task 8

B - Analysis Task 8

C - Summary Task 9

How Phase Equilibrium and Reaction Kinetics Models are Connected and What This Means 9

4. REFERENCES 10

5. APPENDICES 11

TASK 1

A - Mathematical Task

$$x_1 P_{v,1} = y_1 \phi_1 P, \quad (1.1)$$
$$x_2 H_{21} = y_2 \phi_2 P. \quad (1.2)$$

$$\sum_i y_i = \ y_1 + y_2 = 1, \quad (2.1)$$
$$\sum_i x_i = \ x_1 + x_2 = 1. \quad (2.2)$$

Where, $x_i, y_i, x_1, \ x_2, \ P_{v,1}, H_{21}, \ y_1, \ y_2, \ \phi_1, \ \phi_2$ and P are the molar fraction of component (i) in the liquid, molar fraction of component (i) in the vapour, the molar fraction of water, molar fraction of CO_2 dissolved in water, vapour pressure of water, temperature dependent Henry's constant, molar fraction of water vapour, molar fraction of CO_2 in gas form, the fugacity ratio of vapour phase over liquid phase of water, the fugacity in the vapour phase of CO_2 and pressure respectively.

Equations 1 & 2 are a rearrangement of equations 1.1 and 1.2 (1)(2) such that the coefficients (of x_i and y_i) and the right-hand side of the linear equations (1&2) below are more efficiently translated into the coefficient matrix (A) and the known 4x1 array (M_r) respectively. This is done so that $A *$ the linear matrix system of x_i and y_i $(M_{xy}) = M_r$ (in equation 3) is equivalent to the simultaneous (3) system of 4 linear equations (otherwise known as equations 1, 2, 2.1 and 2.2 (4)) solved in matrix form.

$$x_1 * \ P_{v,1} - y_1 * \phi_1 * P = 0 \quad (1)$$
$$x_2 * \ H_{21} - \ y_2 * \phi_2 * P = 0 \quad (2)$$

In order to find values of x_i and y_i as a linear matrix system M_{xy}, a 4x4 coefficient matrix (A) must be synthesised using the coefficients of x_1, x_2, y_1 and y_2 from equations 1, 2, 2.1 and 2.2 (1)(4) (see equation 3). From there, multiplying the inverse of the coefficient matrix (A^{-1}), by the known array (M_r), reveals the values of the matrix M_{xy} as calculated in equation 4. See appendix [i] and appendix [ii] for the full coefficient matrix inversion process (A^{-1}) and the code for Task 1A respectively.

$$A * M_{xy} = M_r$$

$$\therefore \begin{bmatrix} P_{v,1} & 0 & -\phi_1 P & 0 \\ 0 & H_{21} & 0 & -\phi_2 P \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} * \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 P_{v,1} - y_1 \phi_1 P \\ x_2 H_{21} - \ y_2 \phi_2 P \\ x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \quad (3)$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} P_{v,1} & 0 & -\phi_1 P & 0 \\ 0 & H_{21} & 0 & -\phi_2 P \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}^{-1} * \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{-\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{-\phi_1}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{H_{21}\phi_1}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{-P\phi_1\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} \\ \frac{\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{\phi_1}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{-P_{v,1}\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{P\phi_1\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} \\ \frac{-H_{21}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{-P_{v,1}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{P_{v,1}H_{21}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{-P_{v,1}\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} \\ \frac{H_{21}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{P_{v,1}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{-P_{v,1}H_{21}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{H_{21}\phi_1}{H_{21}\phi_1 - P_{v,1}\phi_2} \end{bmatrix} * \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{H_{21}\phi_1 - P\phi_1\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} \\ \frac{P\phi_1\phi_2 - P_{v,1}\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} \\ \frac{P_{v,1}H_{21} - P_{v,1}\phi_2 P}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} \\ \frac{H_{21}\phi_1 P - P_{v,1}H_{21}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} \end{bmatrix} \quad (4)$$

Where, A, M_{xy} and M_r are the coefficient matrix, the linear matrix system of x_i and y_i values and the 4x1 matrix of the right-hand sides of equations 1, 2, 2.1 and 2.2 respectively.

If $A * M_{xy} = M_r$ is true in equation 3, then $M_{xy} = M_r * A^{-1}$ in equation 4 & 5 is also true considering a matrix multiplied by its inverse is the identity matrix (3). When the inverse of A (A^{-1}) is expanded to be $\frac{1}{\det(A)} * adj(A)$ (as seen in equation 5), A^{-1} becomes infinity as $\det(A)$ approaches 0 which is expanded on in equation 5 and used to inform a discussion on the implications of this condition in a real-life system.

Implications of a Condition Where the Determinant of the Coefficient Matrix Becomes Zero

$$M_{xy} = M_r * A^{-1} = \ M_r * \frac{1}{\det(A)} * \mathrm{adj}(A)$$
$$\therefore \det(A) = \ P(H_{21}\phi_1 - P_{v,1}\phi_2) \ = 0 \quad (5)$$
$$\therefore P = 0 \quad \text{and/or} \quad H_{21}\phi_1 - P_{v,1}\phi_2 = 0$$

Where $\det(A)$, and $\mathrm{adj}(A)$ are the determinant of A and the adjoint of A respectively.

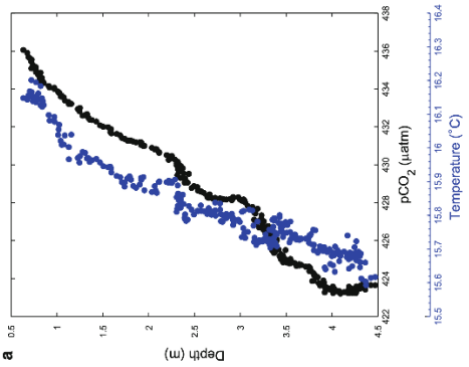
$$H_{21}\phi_1 - P_{v,1}\phi_2 = 0 \ \therefore \ \frac{P_{v,1}}{H_{21}} = \frac{\phi_1}{\phi_2} \text{ also } \frac{x_1 P_{v,1}}{x_2 H_{21}} = \frac{y_1\phi_1}{y_2\phi_2} \therefore \ \frac{x_1}{x_2} = \frac{y_1}{y_2} = 1 \therefore x_1 = x_2 \ \& \ y_1 = y_2 \qquad (6)$$

Equation 6 shows how dividing equations 1.1 and 1.2 ($\frac{x_1 P_{v,1}}{x_2 H_{21}} = \frac{y_1\phi_1}{y_2\phi_2}$)(1)(2)) can be compared to the condition's rearranged formula, det(A) = 0 ($\frac{P_{v,1}}{H_{21}} = \frac{\phi_1}{\phi_2}$), to discern that x_1 and y_1 are x_2 and y_2. Analogously, equations 5 presupposes that pressure (P) and/or $H_{21}\phi_1 - P_{v,1}\phi_2$ are equal to zero (see specifically line 3 of equation 5). The condition also implies $M_{xy} = \infty$, as M_{xy} is $\frac{1}{0} * adj(A)M_r$ under the condition and any number divided by 0 is ∞. Therefore, the three theoretical implications of the condition are as follows: $M_{xy} = \infty, P = 0$ and/or $x_1 = x_2 \ \& \ y_1 = y_2$.

It's worth noting that the condition itself is somewhat unrealistic, firstly because a pressure of 0 in these circumstances is not within reasonable expectations of reality (see task 2C). Secondly, the main assumption of these calculations is that the principal formulas conform to ideal gas constraints (1), therefore, any conclusions should not be considered for values exceeding 1MPa (1).

Concerning a scenario where the pressure (P) of the modelled system approaches 0 and air density isn't constant, a decrease in P may predicate air particles moving apart (therefore moving faster), thus temperature also increases (5). Therefore, the implications of this condition -may imply an increase in anthropogenic CO_2 due to the greenhouse effect (6). Essentially, this phenomenon could indicate that the effects of global warming have reached a concerning momentum, reminiscent of the 32% increase in CO_2 from 280-370 µatm between 1880 and 2000 (7)(1). Additionally, as pressure approaches 0, solubility of CO_2 in water also approaches 0, exacerbating the overabundance of CO_2 in the atmosphere as less CO_2 is absorbed by the ocean (1)

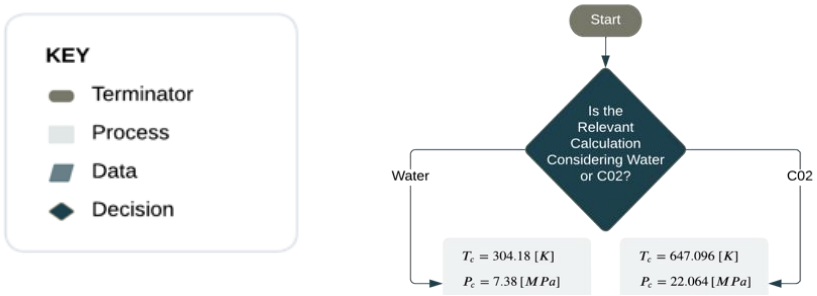
Figure 1. Left: In situ profiles of temperature and pCO2. (8)

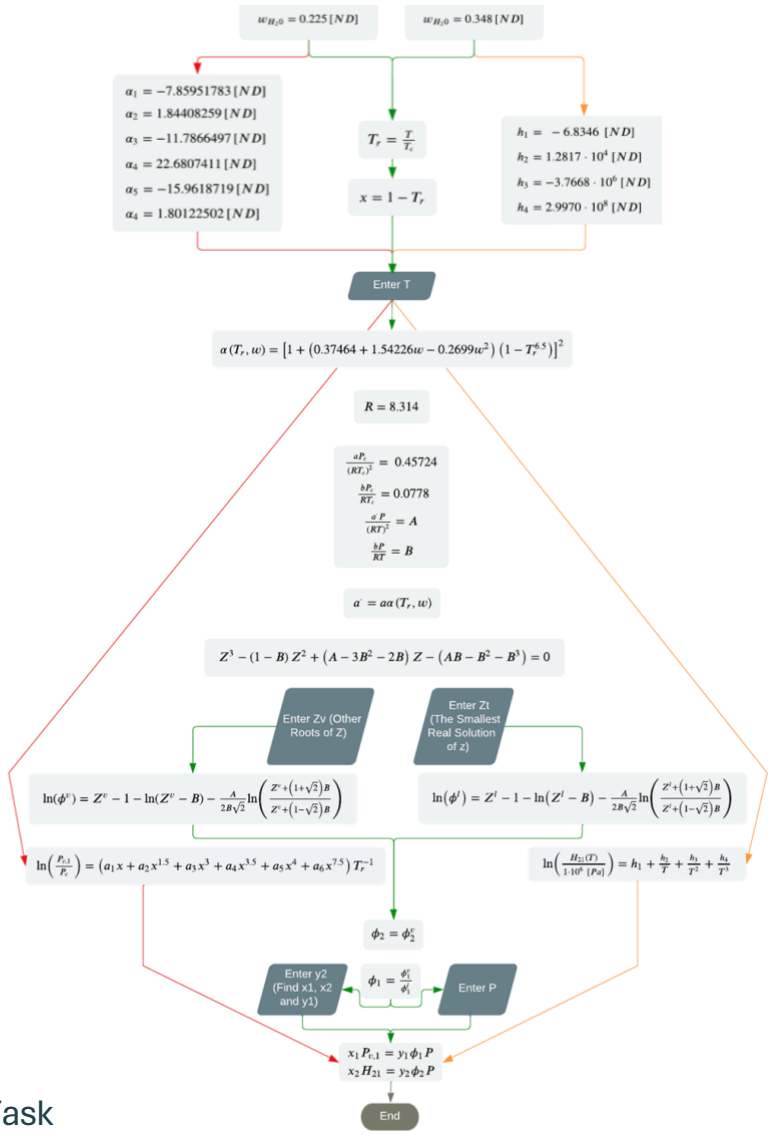


As can be seen in figure 1 above, the partial pressure of CO_2 in water (x_2P = 436 µatm)(8)(1) near the water's surface (0.5 meters deep) is (about 100 µatm) greater than the partial pressure of CO_2 in air (y_2P = 370 µatm)(8)(7)(1). This creates a diffusion gradient such that when $x_1 = x_2 \ \& \ y_1 = y_2$ (equation 6), atmospheric CO_2 rises and oceanic diffusion of CO_2 increases (7). The implication of this could be anything from an increase in x_2P, to an estimated 14-30% reduction in calcification by 2100 (9)(7) due to a decrease in chlorophyll concentration per algal cell and the loss of zooxanthellae (symbiotic algae) within reef ecosystems (especially the Great Barrier Reef (GBR)) (7). Additionally, as M_{xy} tends to ∞, so does $x_1, x_2, y_1 \ \& \ y_2$, this increases the effect of ocean acidification, exacerbates the equilibration of dissolved and atmospheric CO_2 (see task 2B) and may intensify the effects of global warming (10) However, due to the impossibility of infinite values of M_{xy}, environmental planetary self-correction may occur after human involvement ceases therefore damage may not increase exponentially.

B - Communication Task

Figure 2. Schematic of Equations Used in Subsequent Task 1-3 Calculations (1)(4)(11)





C – Modelling Task

The atmosphere is a mixture of many constituent gasses, most of which were not included in the calculations above (equations 1.1 to 2.2). As nitrogen is by far the biggest contributor to the atmosphere's composition, with 3,950 x 10^18 mg of nitrogen being present since 1978, (12) it was chosen as the atmospheric gas component used in the equation 9 extended coefficient matrix M_{xy}.

Carroll, Slupsky and Mather's 1991 article (1) describes a model (in equations 1.1-1.2) that can be extrapolated to facilitate the addition of nitrogen. This is achieved by adding the mass fraction of nitrogen dissolved in water (x_3) and the mass fraction of nitrogen gas (y_3) to equations 2.1-2.2 (4)which become equations 7 & 8). Similarly, equation 1.3 is an extension of equation 2 where the coefficient of x_3 (H_{321}) is Henry's constant for the system H_2O + CO_2 + N_2(1), just as ϕ_3 and P are values of fugacity and pressure true for this extended model. (see appendix[iii] for the respective code).

$$x_3 * H_{321} - y_3 * \phi_3 * P = 0 \tag{1.3}$$

$$\sum_i y_i = y_1 + y_2 + y_3 = 1, \tag{7}$$

$$\sum_i x_i = x_1 + x_2 + x_3 = 1. \tag{8}$$

$$A * M_{xy} = M_r \therefore \begin{bmatrix} P_{v,1} & 0 & 0 & -\phi_1 P & 0 & 0 \\ 0 & H_{321} & 0 & 0 & -\phi_2 P & 0 \\ 0 & 0 & H_{321} & 0 & 0 & -\phi_3 P \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} * \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} x_1 P_{v,1} - y_1 \phi_1 P \\ x_2 H_{321} - y_2 \phi_2 P \\ x_3 H_{321} - y_3 \phi_3 P \\ x_1 + x_2 + x_3 \\ y_1 + y_2 + y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \tag{9}$$

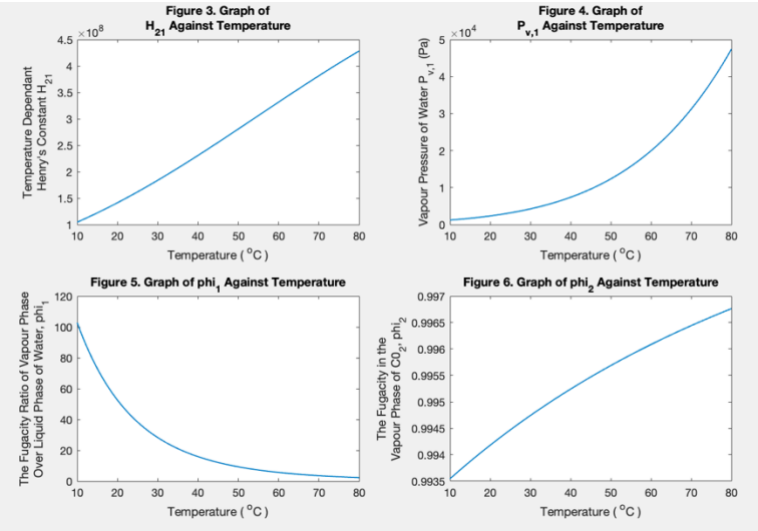
The quantities required are therefore the values of x_3 , ϕ_3, and y_3 for which nitrogen is the molar fraction, fugacity and gaseous molar fraction. Computationally, a simultaneous-model of equations 1.1-2.2 can be extended to represent the properties of nitrogen, where equations 1.3, 6 and 7 can be applied to the coefficient matrix in task 1A to form a 5x6 matrix of all three elements (water, CO_2 and nitrogen as in equation 9).

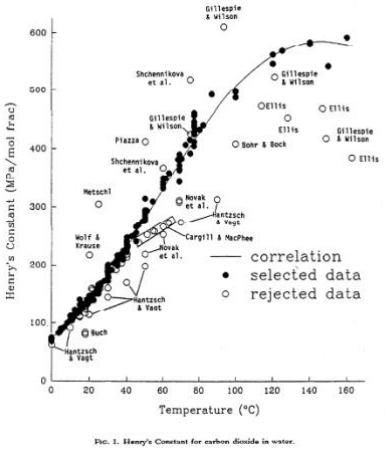
TASK 2

A - Coding Task

The code for Task 2A (in appendix[iv]) establishes the function file (Task2_Function_Values.mlx) where the function zeros(length(T_K),4) is used such that each coefficient of the cubic equation (11) in the calculation of phi 1and 2 is a collum in an array the length of the temperature datapoints (∴ it can be reused in task 2B & 2C). A for loop is implemented solely as a workaround to the roots() function (13) which can only process equations one at a time (roots are filtered to be real and positive).

Figure 3, 4, 5, 6 and 7. Task 2A MATLAB Graphs (1) and figure 7. Henry's Constant for Carbon Dioxide in Water

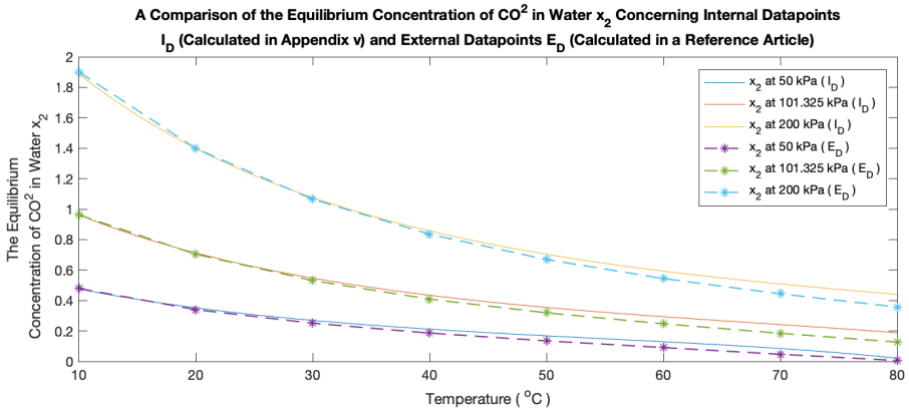




As for assertions that can be made when considering the graphs above, (constants H_{21} and $P_{v,1}$ are both essentially functions of temperature, therefore are under less scrutiny). Given that ϕ_1 is $\frac{\phi_1{}^{v}}{\phi_1{}^{l}}$ (vapour over liquid fugacity for H_20) and ϕ_2 is $\phi_2{}^{v}$ (fugacity in vapour phase for CO_2), ϕ_1 is always a somewhat large number over 1 while ϕ_2 must be less that 1. The graphs (figures 3-6) above seem to follow a logical correlation which can be seen in the similarities between figures 3 and 7 (measuring H_{21} fluctuation across time). The consistency in H_{21}results across Piazza, Metschl, Ellis and others' work is likely due to the simple, repeatable nature of the constant (1).

B - Validation Task

Figure 8. Task 2B graph



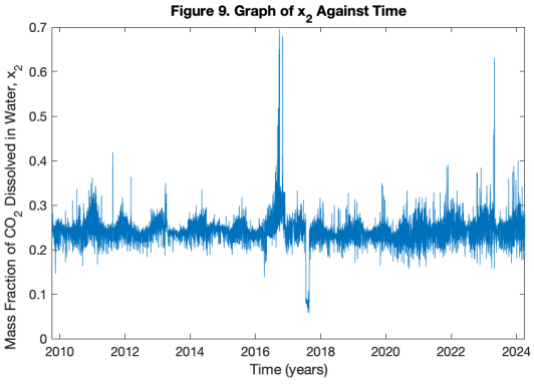
The line graphs in figure 8 are a calculation of values of x_2 over temperature with respect to 3 given pressures (50 kPa, 101.325 kPa, 200 kPa). Values of x_2 are calculated using the values found in

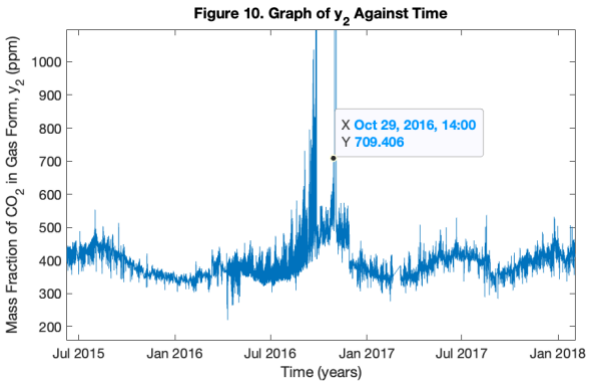
equation 4 where $\begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \frac{H_{21}\phi_1 - P\phi_1\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} \\ \frac{P\phi_1\phi_2 - P_{v,1}\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} \\ \frac{P_{v,1}H_{21} - P_{v,1}\phi_2 P}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} \\ \frac{H_{21}\phi_1 P - P_{v,1}H_{21}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} \end{bmatrix}$ ($\therefore x_2 = \frac{P\phi_1\phi_2 - P_{v,1}\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2}$) and are compared to the calculated solubilities for CO_2 in water as seen in the 1991 article (1). The task 2B code found in appendix[v] also uses MATLAB documentation(14) to create a table of these figures for a direct comparison with its counterpart in the 1991 Carroll, Slupsky and Mather article (1).

The comparison of my values and Carroll, Slupsky and Mather's reveal a striking similarity in our data. The minor deviations at low and high temperatures may be a result of using different calculation methods where the integration of Peng-Robinson's cubic equation into the internal calculations (I_D) could have created a more exaggerated curvilinear shape than their external data (E_D). In relation to this, the largely negative correlation between x_2 and temperature in both scenarios supports my point in task 1A, where the increase in atmospheric and oceanic CO_2 equilibration means the loss of the earth's natural ability to store CO_2 gas. In other words, the increase in temperature due to global warming would mean lower value of x_2 which is a result of equilibration decreasing the diffusion gradient from air to water. This would only exacerbate the ongoing reef damage at the hands of ocean acidification and the greater effects of global warming expanded upon in this paper.

C - Modelling Task

Graphs 8 and 9 oscillate seasonally at a similar time period (≈ a year) where y_2 in figure 10 decreases in early spring and increases the rest of the year(15). Conversely, in figure 9, values of x_2 decrease in summer and increase in winter such that the two graphs are out of phase(16). For figure 9, the negative correlation of x_2 and temperature (in figure 8 above) sets the premise that when it is colder (in winter) values of x_2 increase (and vice versa for summer). Furthermore, the amplitude of figure 9's oscillations increase with time. This is an unfortunate effect of x_2P (oceanic partial pressure of CO_2) increasing due to anthropogenic emissions which has increased the amplitude of figure 9 by 2.2 ∓ 0.4 µatm decade^{-1} from 1982-2015 poleward of 10 degree latitude(17)(16). Alternatively, the unique seasonal patterning of figure 10 can be attributed to decreasing photosynthesis rates as vegetation grows in spring (therefore exchanging less CO_2 for O_2)(15)(18). The graphs below concern the GBR specifically however the analysis above considers this a microcosm of general oceanic trends.





Concerning the MATLAB code for task 2C (see appendix [vi]) the data from the brief (2) had to be transformed into a manipulable array using MATLAB's import data tool. From there, the GBR datapoints are isolated for a more precise analysis. The function sortrows() is used to order data sequentially while function fillmissing() and the code (y_2(y_2>1200) = 709 (for each extra line graph in the code) cleans the data to fill missing values and remove egregious outliers. Interestingly enough, one of these line plots is pressure against time, where the sinusoidal nature of the correlation may confirm P = 0 is a unrealistic condition given pressure is unlikely to be a purely constant value (due to the complexity of natural systems). The code also rearranges equation 1.2 (1) into $x_2 = \frac{y_2\phi_2 P}{H_{21}}$ to solve for x_2.

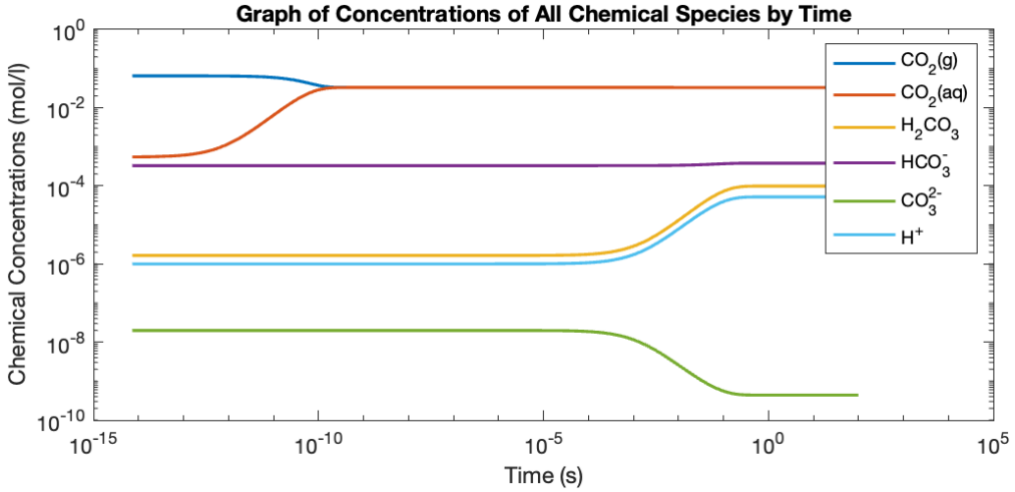
TASK 3

A - Coding Task

The MATLAB code in task 3A (found in appendix [vii]) uses the ode15s function because the large reaction rate variation necessitates a stiff solver (19). It also uses as an inbuilt function file to store the 6 chemical concentrations (mol/liter) from the brief (2). This output, called ODEsys_reaction (or dydx), is then imputed into the ode15s function so that the complex system of 6 non-linear differential equations can be integrated, producing an array of the individual concentrations (y_con_array) as seen in figure 11.

Additionally, the loglog() function (20) is used to achieve graph readability in figure 11, where absolute and relative tolerances are set to 1e-12 and 1e-6 respectively to account for the small initial conditions (that can be $\leq 10^{-7}$).

Figure 11. Graph of Concentrations of All Chemical Species by Time



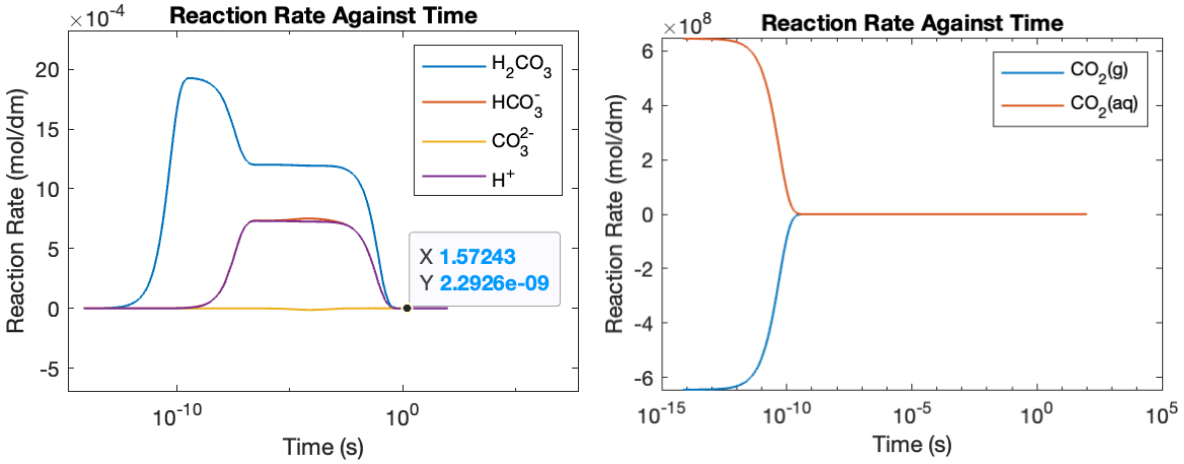
As is seen in the chain reactions of figure 11, when CO_2 dissolves into the ocean, it indirectly introduces H_2CO_3(carbonic acid), $HCO_3{}^-$(bicarbonate ions), $CO_3{}^{2-}$ (carbonate ions))and H^-(hydrogen ions) into the oceanic climate. Under conditions where the system's CO_2 input is low, the production of carbonate and bicarbonate ions (see reaction R4 in the brief (2)) use up carbonic acid so that the presence of $CO_3{}^{2-}$and $HCO_3{}^-$ act as natural balancing systems. However, in the current maritime climate, the decreased $CO_3{}^{2-}$ production in relation to H_2CO_3 may tip the scales towards an overproduction of H_2CO_3(21) which could have effects such as damaging the calcium-carbonate shell and skeleton building organisms that build corals (such as the GBR)(22).
Similarly, the production of H^- from dissolved CO_2 (see green line in figure 11) has substantially decreased the oceans PH from 8.2 to 8.07, (9)(21) such that concentration of H^- has increased by 30%(21).

B - Analysis Task

The constants from Table 3 and the numerical solution (y_con_array) of the system 5.1-5.6 from the brief (2) were used to code the semi-logarithmic graph below (figure 12) (see appendix [viii] for the task 3B code). This was done using the function semilogx() which plots a logarithmic x-axis and a linear y-axis (unlike in task 3A where both axes needed logarithmic filters).
The chemical components of reactions R1-4(2) have been divided into two graphs (in figure 12 below) where the leftmost graph contains chemicals of reaction rates from about -1e-5 mol/dm to 20e-4 mol/dm. The rightmost graph, therefore, contains the chemicals with reaction rates between -6e8 and 6e8 mol/dm (atmospheric and dissolved CO_2). These compounds have an equal and opposite rate of reaction because they act in direct, isolated cause and effect of each other (the speed of the CO_2 reactions could be because CO_2 is extremely water-soluble). Comparatively, the production rate of H_2CO_3, $HCO_3{}^-$, $CO_3{}^{2-}$and H^- are over 1e12 times smaller, where H_2CO_3, $HCO_3{}^-$and H^- have similar

reaction rates and $CO_3{}^{2-}$ is noticeably slower. A possible reason for this could be that the compounds needed to complete the reactions are harder to come by, or that a lot of energy is required for the reaction to take place.
The principal estimation of the time the system takes to reach steady state equilibrium is 1.5 s (1d.p.) (labelled on the leftmost graph in figure 12) which is calculated by finding the point at which the reaction stops.

Figure 12. Reaction Rates of Chemical Compounds Against Time



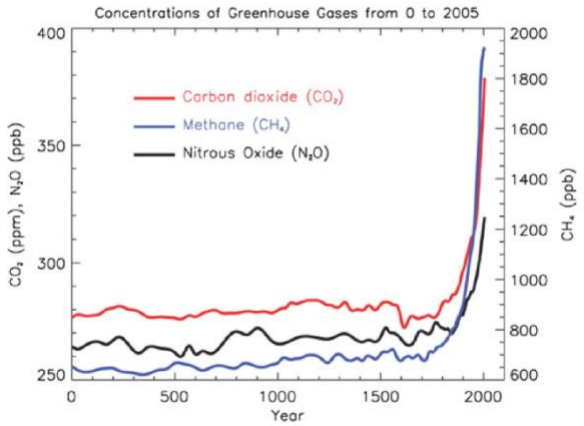
C - Summary Task

The reaction kinetics model seems to mainly consider specifics regarding the effects of $HCO_3{}^-$, $CO_3{}^{2-}$, H^- and H_2CO_3 in a way that sheds light on the consequences of global warming at the chemical (ionic) level. Conversely, the phase equilibrium models, (tasks 1 & 2) are different in that they tend to be subordinate to overall trends. In summary, using both models to analyse a topic brings into consideration both macro and microscopic perspectives (which is unlike if they were to be used in isolation). The limits of one model are the strengths of the other and vice versa.
The macroscopic outlook of a phase equilibrium model therefore presupposes that the point of data analysis is to describe the current state of the world (e.g. the positive and negative correlations in the task 2A graphs), or extrapolate on current data patterns. Whereas the reaction kinetics model is able to dissect and understand the particulars of a phenomenon, and therefore predict how patterns (e.g. the chemical reactions caused by CO_2 in the ocean) may change in the future, not just how they currently operate.

How Phase Equilibrium and Reaction Kinetics Models are Connected and What This Means

As touched upon in task 3A, the kinetics model finds that the rapid chemical conversion of CO_2 into H^- since the 18th century, has decreased the PH of the ocean from 8.2 to 8.07, (9)(21) causing ocean acidification. Here, information about the increase in CO_2 (x_2) due to global warming (from figure 13(8)) can be applied to the kinetic model's knowledge that x_2 is negatively correlated to temperature (task 2B). Therefore ascertaining that the increase in the chemical production of dissolved CO_2, (from atmospheric CO_2) is due to the warming of our globe.

Figure 13. Concentrations of Greenhouse Gases from 0 to 2005(8)



4. REFERENCES

1. Carroll JJ, Slupsky JD, Mather AE. The Solubility of Carbon Dioxide in Water at Low Pressure. J Phys Chem Ref Data. 1991 Nov 1;20(6):1201–9.

2. ENGF0003 Assessment 2 Brief - UCL Mathematical Modelling and Analysis. CO 2 Dissolution in Water [Internet]. 2024 [cited 2025 Jan 2]. Available from: https://ucl-eu-west-2-moodle-sitedata.s3.eu-west-2.amazonaws.com/a6/a0/a6a0a7d33d8eefefd48e7026929af38607e93b29?response-content-disposition=inline%3B%20filename%3D%22ENGF0003_CW_24-25_F.pdf%22&response-content-type=application%2Fpdf&X-Amz-Content-Sha256=UNSIGNED-PAYLOAD&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIA47YHZF637GKGWUJC%2F20250102%2Feu-west-2%2Fs3%2Faws4_request&X-Amz-Date=20250102T152128Z&X-Amz-SignedHeaders=host&X-Amz-Expires=21572&X-Amz-Signature=ebac19aa1a88f2f687fbcbe635c597ba731b8decc89b74a8b27f2ba5ae49e6bb

3. Earl R. Week 1-Vectors and Matrices University of Oxford [Internet]. 2003 [cited 2025 Jan 3]. Available from: https://www.maths.ox.ac.uk/system/files/attachments/matrices-3.pdf

4. Carey VP. The properties of gases & liquids. Exp Therm Fluid Sci. 1988 Oct;1(4):409.

5. Kakasuleff K. NASA - Convection Activities. [cited 2025 Jan 4]; Available from: https://www.grc.nasa.gov/www/k-12/Summer_Training/FranktonES/Convection_main_page.html

6. Whitten C. Robert. GREENHOUSE GASES AND GLOBAL WARMING. Naval War College Review [Internet]. 2003 [cited 2025 Jan 4];56:142–5. Available from: https://www.jstor.org/stable/26393960

7. Reynaud S, Leclercq N, Romaine-Lioud S, Ferrier-Pagés C, Jaubert J, Gattuso J. Interacting effects of CO_2 partial pressure and temperature on photosynthesis and calcification in a scleractinian coral. Glob Chang Biol. 2003 Nov 9;9(11):1660–8.

8. Arneth A, Arnold S, Baker A, Bakker D, Bange H. Ocean-Atmosphere Interactions of Gases and Particles [Internet]. Liss PS, Johnson MT, editors. Berlin, Heidelberg: Springer Berlin Heidelberg; 2014 [cited 2025 Jan 5]. Available from: file:///Users/samanthadufumulleruri/Downloads/1001934.pdf

9. Gattuso JP, Lavigne H. Technical Note: Approaches and software tools to investigate the impact of ocean acidification. Biogeosciences. 2009 Oct 8;6(10):2121–33.

10. Anthony KRN, A. Kleypas J, Gattuso J. Coral reefs modify their seawater carbon chemistry – implications for impacts of ocean acidification. Glob Chang Biol. 2011 Dec 6;17(12):3655–66.

11. Peng DY, Robinson DB. A New Two-Constant Equation of State. Industrial & Engineering Chemistry Fundamentals. 1976 Feb 1;15(1):59–64.

12. Holland D. Heinrich. Release and Recycling of Volatiles during Earth History. In: The Chemical Evolution of the Atmosphere and Oceans. Princeton University Press; 2020. p. 64–88.

13. MathWorks. Roots - MATLAB Documentation [Internet]. [cited 2025 Jan 5]. Available from: https://uk.mathworks.com/help/matlab/ref/roots.html

14. MathWorks. Tables - MATLAB Documentation [Internet]. [cited 2025 Jan 5]. Available from: https://uk.mathworks.com/help/matlab/tables.html

15. US Environmental Protection Agency. EPA - Carbon Through The Seasons [Internet]. [cited 2025 Jan 6]. Available from: https://archive.epa.gov/climatechange/kids/documents/carbon-through-the-seasons.pdf

16. Landschützer P, Gruber N, Bakker DCE, Stemmler I, Six KD. Strengthening seasonal marine CO2 variations due to increasing atmospheric CO2. Nat Clim Chang. 2018 Feb 29;8(2):146–50.

17. Rodgers KB, Sarmiento JL, Aumont O, Crevoisier C, de Boyer Montégut C, Metzl N. A wintertime uptake window for anthropogenic CO_2 in the North Pacific. Global Biogeochem Cycles. 2008 Jun 31;22(2).

18. Lindsey R. Climate Change: Atmospheric Carbon Dioxide [Internet]. 2024 [cited 2025 Jan 6]. Available from: https://www.climate.gov/news-features/understanding-climate/climate-change-atmospheric-carbon-dioxide#:~:text=The%20seasonal%20cycle%20of%20highs,is%20driven%20by%20human%20activities.

19. MathWorks. Ode15s - MATLAB Documentation [Internet]. [cited 2025 Jan 6]. Available from: https://uk.mathworks.com/help/matlab/ref/ode15s.html?searchHighlight=ode15s&s_tid=srchtitle_support_results_1_ode15s

20. MathWorks. Loglog - MATLAB Documentation [Internet]. [cited 2025 Jan 6]. Available from: https://uk.mathworks.com/help/matlab/ref/loglog.html

21. Logan CA. A Review of Ocean Acidification and America's Response. Bioscience. 2010 Nov;60(10):819–28.

22. Strong AL, Kroeker KJ, Teneva LT, Mease LA, Kelly RP. Ocean Acidification 2.0: Managing our Changing Coastal Ocean Chemistry. Bioscience. 2014 Jul 1;64(7):581–92.

5. APPENDICES

[i] Step By Step Coefficient Matrix Inversion :

$$A = \begin{bmatrix} P_{v,1} & 0 & -\phi_1 P & 0 \\ 0 & H_{21} & 0 & -\phi_2 P \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad \text{STEP 1}$$

$$\therefore A^{-1} = \frac{1}{\det(A)} * adj(A)$$

Calculating the Determinant ($\det(A)$) :

$$\det(A) = P_{v,1} * \begin{vmatrix} H_{21} & 0 & -\phi_2 P \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} - \phi_1 P * \begin{vmatrix} 0 & H_{21} & -\phi_2 P \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\det(A) = P_{v,1} * \Big(H_{21} * (0-0) - 0 * (1-0) - \phi_2 P * (1-0) \Big) - \phi_1 P * \big(0 * (1-0) - H_{21} * (1-0) - \phi_2 P * (0-0) \big) \quad \text{STEP 2}$$

$$\det(A) = P_{v,1} * \big(-\phi_2 P \big) - \phi_1 P * (-H_{21})$$

$$\det(A) = P(H_{21}\phi_1 - P_{v,1}\,\phi_2)$$

Calculating the Adjoint ($adj(A)$) :

$$adj(A) = C^T$$

$$= \begin{bmatrix} +\begin{vmatrix} H_{21} & 0 & -\phi_2 P \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} & -\begin{vmatrix} 0 & 0 & -\phi_2 P \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} & +\begin{vmatrix} 0 & H_{21} & -\phi_2 P \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} & -\begin{vmatrix} 0 & H_{21} & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \\ -\begin{vmatrix} 0 & -\phi_1 P & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} & +\begin{vmatrix} P_{v,1} & -\phi_1 P & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} & -\begin{vmatrix} P_{v,1} & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} & +\begin{vmatrix} P_{v,1} & 0 & -\phi_1 P \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \\ +\begin{vmatrix} 0 & -\phi_1 P & 0 \\ H_{21} & 0 & -\phi_2 P \\ 0 & 1 & 1 \end{vmatrix} & -\begin{vmatrix} P_{v,1} & -\phi_1 P & 0 \\ 0 & 0 & -\phi_2 P \\ 0 & 1 & 1 \end{vmatrix} & +\begin{vmatrix} P_{v,1} & 0 & 0 \\ 0 & H_{21} & -\phi_2 P \\ 0 & 0 & 1 \end{vmatrix} & -\begin{vmatrix} P_{v,1} & 0 & -\phi_1 P \\ 0 & H_{21} & 0 \\ 0 & 0 & 1 \end{vmatrix} \\ -\begin{vmatrix} 0 & -\phi_1 P & 0 \\ H_{21} & 0 & -\phi_2 P \\ 1 & 0 & 0 \end{vmatrix} & +\begin{vmatrix} P_{v,1} & -\phi_1 P & 0 \\ 0 & 0 & -\phi_2 P \\ 1 & 0 & 0 \end{vmatrix} & -\begin{vmatrix} P_{v,1} & 0 & 0 \\ 0 & H_{21} & -\phi_2 P \\ 1 & 1 & 0 \end{vmatrix} & +\begin{vmatrix} P_{v,1} & 0 & -\phi_1 P \\ 0 & H_{21} & 0 \\ 1 & 1 & 0 \end{vmatrix} \end{bmatrix}^T$$

STEP 3

$$\therefore adj(A) = \begin{bmatrix} -\phi_2 P & -\phi_1 P & -\phi_1 P H_{21} & -\phi_1 \phi_2 P^2 \\ \phi_2 P & \phi_1 P & -\phi_2 P P_{v,1} & \phi_1 \phi_2 P^2 \\ -H_{21} & -P_{v,1} & H_{21} P_{v,1} & -\phi_2 P P_{v,1} \\ H_{21} & P_{v,1} & -H_{21} P_{v,1} & \phi_1 P H_{21} \end{bmatrix}$$

The Inverse of the Coefficient Matrix (A^{-1}) :

$$A^{-1} = \frac{1}{\det(A)} * adj(A)$$

$$A^{-1} = \frac{1}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} \begin{bmatrix} -\phi_2 P & -\phi_1 P & -\phi_1 P H_{21} & -\phi_1 \phi_2 P^2 \\ \phi_2 P & \phi_1 P & -\phi_2 P P_{v,1} & \phi_1 \phi_2 P^2 \\ -H_{21} & -P_{v,1} & H_{21} P_{v,1} & -\phi_2 P P_{v,1} \\ H_{21} & P_{v,1} & -H_{21} P_{v,1} & \phi_1 P H_{21} \end{bmatrix}$$

STEP 4

$$A^{-1} = \begin{bmatrix} \frac{-\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{-\phi_1}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{H_{21}\phi_1}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{-P\phi_1\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} \\ \frac{\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{\phi_1}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{-P_{v,1}\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} & \frac{P\phi_1\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} \\ \frac{-H_{21}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{-P_{v,1}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{P_{v,1}H_{21}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{-P_{v,1}\phi_2}{H_{21}\phi_1 - P_{v,1}\phi_2} \\ \frac{H_{21}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{P_{v,1}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{-P_{v,1}H_{21}}{P(H_{21}\phi_1 - P_{v,1}\phi_2)} & \frac{H_{21}\phi_1}{H_{21}\phi_1 - P_{v,1}\phi_2} \end{bmatrix}$$

Where $\det(A)$, $adj(A)$ and C^T are ,,,, and ... respectively.

[ii] MATLAB Code for Task 1A:

```
% TASK 1A

% Expressing Equations as a Linear Matrix System:
% syms function allows you to mathematically compute undefined variables

syms x_1 x_2 y_1 y_2 phi_1 phi_2 P_v1 P H_21
A = [P_v1 0 -phi_1*P 0; 0 H_21 0 -phi_2*P; 1 1 0 0; 0 0 1 1]
```

```
M_r = [0 0 1 1]'
M_xy = (A^-1) * M_r
```

[iii] MATLAB Code for Task 1C:

```
% TASK 1C

% Analysing Under Which Conditions This Linear Matrix System is Invalid:

syms x_3 y_3 phi_3 H_321
A_2 = [P_v1 0 0 -phi_1*P 0 0; 0 H_321 0 0 -phi_2*P 0;0 0 H_321 0 0 -phi_3...
    *P; 1 1 1 0 0 0; 0 0 0 1 1 1]
M_r2_knowns = [0 0 0 1 1]'
M_r2_unknowns = [(x_1*P_v1)-(y_1*phi_1*P);(x_2*H_321)-(y_2*phi_2*P);(x_3*...
    H_321)-(y_3*phi_3*P);x_1+x_2+x_3;y_1+y_2+y_3]
M_xy2 = [x_1; x_2; x_3; y_1; y_2; y_3]
```

[iv] MATLAB Task 2 Function File (The code for 2A is directly below):

```
% Below is the Code of a Function File (Task2_Function_Values) that
% Calculates H_21, P_v1, phi_1 & phi_2:

% Where:

% H_21 = Temperature-dependent Henry's constant
% P_v1 = Vapour pressure of water
% phi_1 = Fugacity ratio of vapour phase over liquid phase of water
% phi_2 = Fugacity in the vapour phase of CO2
% P = Pressure (Pa)
% T_K = Temperature (K)

function [H_21, P_v1, phi_1, phi_2] = Task2_Function_Values(P, T_K)

% All variables are expressed in their standard units

% Defining Constants and Variables (Regarding Water and CO2):
% Where column 1 (:,1) of the arrays below (Tr, Pc (critical pressure) &
% wh_af (acentric factor)) concern CO2 just as column 2 (:,2) of the
% arrays below concern water.

Tc = [304.18, 647.096]
Pc = [7.38*10^6, 22.064*10^6]
```

```
wh_af = [0.225, 0.348]

% Calculating Values of H_21
h1 = -6.8346
h2 = 1.2817*10^4
h3 = -3.7668*10^6
h4 = 2.9970*10^8

H_21 = (1*10^6).*exp((h1+(h2./T_K)+(h3./T_K.^2)+(h4./T_K.^3)));

% Calculating Values of P_v1
a1 = -7.85951783
a2 = 1.84408259
a3 = -11.7866497
a4 = 22.6807411
a5 = -15.9618719
a6 = 1.80122502

% Store values of Tr & x as an array
Tr = T_K./Tc;
x = 1 - Tr;

% Since P_v1 is the vapour pressure of water, the variables below (Pc, x &
% Tr) pertain to water (therefore only the second column of these variables
% are used (e.g. x(:,2)).
P_v1 = Pc(:,2).*exp((a1.*x(:,2)+a2.*x(:,2).^1.5+a3.*x(:,2).^3+a4.*x(:,2) ...
    .^3.5+a5.*x(:,2).^4+a6.*x(:,2).^7.5).*Tr(:,2).^(-1));

% Below is the Beginning of the Process to Calculate Values of phi_1:
R = 8.314
alpha = (1+(0.37464+1.54226.*(wh_af)-0.2699.*(wh_af).^2).*(1 ...
    -Tr.^0.5)).^2;

% Note that column 1 (:,1) of the arrays below concern CO2 just as column
% 2 (:,2) of the arrays below concern water.
a = (0.45724.*((R.*Tc).^2))./Pc
a_star = a.*alpha
b = (0.0778.*(R.*Tc))./Pc
A = (a_star.*P)./((R.*T_K).^2)
B = (b.*P)./(R.*T_K)

% Z Peng-Robinson cubic coefficients (for CO2)
Zcoeff_co2 = zeros(length(T_K),4);
Zcoeff_co2(:,1) = 1;
Zcoeff_co2(:,2) = -(1-B(:,1));
Zcoeff_co2(:,3) = A(:,1)-3.*B(:,1).^(2)-2.*B(:,1);
Zcoeff_co2(:,4) = -(A(:,1).*B(:,1)-B(:,1).^(2)-B(:,1).^(3));
```

```
% Z Peng-Robinson cubic coefficients (for water)
Zcoeff_water = zeros(length(T_K),4);
Zcoeff_water(:,1) = 1;
Zcoeff_water(:,2) = -(1-B(:,2));
Zcoeff_water(:,3) = A(:,2)-3.*B(:,2).^(2)-2.*B(:,2);
Zcoeff_water(:,4) = -(A(:,2).*B(:,2)-B(:,2).^(2)-B(:,2).^(3));

% Filtering the roots to get real and positive values (in other words
% values that are valid).

% Finding Real Positive Roots (for C02 and Water):
% Note that for each temperature value, CO2 has only 1 unique real root
% while water has 3. This is why Zroots_co2_real is a 1 column array while
% Zroots_water_real is a 3 column array.
Zroots_co2_real = zeros(length(T_K),1);
Zroots_water_real = zeros(length(T_K),3);
for e = 1:length(T_K)
    Zroots_co2 = roots(Zcoeff_co2(e,:));
    Zroots_co2_real(e,:) = Zroots_co2(imag(Zroots_co2)==0);
    Zroots_water = roots(Zcoeff_water(e,:));
    Zroots_water_real(e,:) = Zroots_water(imag(Zroots_water)==0);
end

% For roots of phi_l (water), values are the smallest real solution of z
% (Zroots_co2_real(:,3)).
% For roots of phi_v (water), largest values of z (Zroots_co2_real(:,1))
% are used. (There is only 1 column/set of values for phi_v (CO2)).

% Calculating phi_v for CO2:
phi_v_co2 = exp(Zroots_co2_real-1-log(Zroots_co2_real-B ...
    (:,1))-(A(:,1)./(B(:,1).*2.*sqrt(2))).*log((Zroots_co2_real+(1+ ...
    sqrt(2)).*B(:,1))./(Zroots_co2_real+(1-sqrt(2)).*B(:,1))));

% Calculating phi_l and phi_v for Water:
phi_l_water = exp(Zroots_water_real(:,3)-1-log(Zroots_water_real(:,3)...
    -B(:,2))-(A(:,2)./(B(:,2).*2.*sqrt(2))).*log((Zroots_water_real(:,3)...
    +(1+sqrt(2)).*B(:,2))./(Zroots_water_real(:,3)+(1-sqrt(2)).*B(:,2))));

phi_v_water = exp(Zroots_water_real(:,1)-1-log(Zroots_water_real(:,1)...
    -B(:,2))-(A(:,2)./(B(:,2).*2.*sqrt(2))).*log((Zroots_water_real(:,1)...
    +(1+sqrt(2)).*B(:,2))./(Zroots_water_real(:,1)+(1-sqrt(2)).*B(:,2))));

% phi_1:
phi_1 = phi_v_water./phi_l_water;

% phi_2:
```

```
phi_2 = phi_v_co2;
```

MATLAB Code for Task 2A:

```
% Task 2A

% clear

% Below, values of pressure and temperature (in Kelvin so that they don't
% need to be converted when input into an equation) from question 2A are
% defined. (The value for 0K (Kelvin) or absolute zero is taken as -273.15
% degrees celcius).
P = 101.325*10^3
T_K = [10:1:80]'+273.15

% Calling the Task 2 Function File Above to Find Values of H_21, P_v1, phi_1
% & phi_2:
[H_21, P_v1, phi_1, phi_2] = Task2_Function_Values(P, T_K)

% Note that pressure is constant in the graphs below therefore the values
% under consideration fluctuate with respect to temperature exclusively.

% Plotting the Figures in a 4x4 Grid

% H_21
figure
subplot(2,2,1)
plot(T_K-273.15,H_21,LineWidth=1)
% Use of xlim better fits the line graph to the axes (so there is no
% discrepancy between the start of the graph and the start of the x axes).
% Additionally, temperature is converted from kelvin to degrees celcius as
% the question suggests.
xlim([T_K(1)-273.15,T_K(end)-273.15])
xlabel("Temperature ( ^oC )")
ylabel("Temperature Dependant Henry's Constant H_2_1 (ND)")
title("Figure ??. Graph of H_2_1 Against Temperature")

% P_v1
hold on
subplot(2,2,2)
plot(T_K-273.15,P_v1(:,1),LineWidth=1)
xlim([T_K(1)-273.15,T_K(end)-273.15])
xlabel("Temperature ( ^oC )")
ylabel("Vapour Pressure of Water P_v_,_1 (Pa)")
title("Figure ??. Graph of P_v_,_1 Against Temperature")
```

```
% phi_1:
subplot(2,2,3)
plot(T_K-273.15,phi_1,LineWidth=1)
xlim([T_K(1)-273.15,T_K(end)-273.15])
xlabel("Temperature ( ^oC )")
ylabel("The Fugacity Ratio of Vapour Phase Over Liquid Phase of" + ...
    " Water, phi_1 (ND?)")
title("Figure ??. Graph of phi_1 Against Temperature")

% phi_2:
subplot(2,2,4)
plot(T_K-273.15,phi_2,LineWidth=1)
xlim([T_K(1)-273.15,T_K(end)-273.15])
xlabel("Temperature ( ^oC )")
ylabel("The Fugacity in the Vapour Phase of C0_2, phi_2 (ND?)")
title("Figure ??. Graph of phi_2 Against Temperature")
```

[v] MATLAB Code for Task 2B:

```
% Task 2B

% clear

T_K = [10:1:80]'+273.15
P_B = [50, 101.325, 200].*10.^3

% The for loop below (using the function file Task2_Function_Values
% calculated in Task 2A (see appendix iv)) calculates x_2 (phi_1 & phi_2)
% for new values of Pressure in array P_B (as values of phi change for
% different pressures).

for h = [1 2 3]
[H_21_2b, P_v1_2b, phi_1_2b(:,h), phi_2_2b(:,h)] = Task2_Function_Values...
    (P_B(h),T_K);
end

% As found in Task 1A x_2 is:
x_2_I_D = ((P_B.*phi_1_2b.*phi_2_2b)-(P_v1_2b.*phi_2_2b))./((H_21_2b ...
    .*phi_1_2b)-(P_v1_2b.*phi_2_2b));

% Variable x_2_E_R is a transposition of the values in table three of the
% external results (E_R) from the 1991 article (Carroll, Slupsky and
% Mather, 1991)(1).
```

```
x_2_E_D(:,1) = [0.479;0.340;0.251;0.186;0.135;0.091;0.047;0.006]
x_2_E_D(:,2) = [0.962;0.704;0.531;0.409;0.319;0.247;0.185;0.127]
x_2_E_D(:,3) = [1.90;1.398;1.065;0.835;0.669;0.544;0.444;0.357]

% The table below calculates the equilibrium concentration of CO2 in water
% x_2 with fluctuations in pressure and temperature (this can be compared
% with table 3 from refence (Carroll, Slupsky and Mather, 1991)(1)):

x2_50kPa = 10^3.*x_2_I_D([1:10:end],1);
x2_101_325kPa = 10^3.*x_2_I_D([1:10:end],2);
x2_200kPa = 10^3.*x_2_I_D([1:10:end],3);
Temperature_Degrees_Celcius = [10:10:80]';

T = table(Temperature_Degrees_Celcius,x2_50kPa,x2_101_325kPa,x2_200kPa)

% The x-axes is altered so that degrees celcius is used (to stay true to
% the specifics of question 2B).
figure
plot(T_K-273.15,x_2_I_D.*10^3,"-")
hold on
plot(Temperature_Degrees_Celcius',x_2_E_D,"--*",LineWidth=1)
xlabel("Temperature ( ^oC )")
ylabel("The Equilibrium Concentration of CO^2 in Water x_2 (ND)")
title("A Comparison of the Equilibrium Concentration of CO^2 in Wat" + ...
    "er x_2 Concerning Internal Datapoints I_D (Calculated in Appen" + ...
    "dix v) and External Datapoints E_D (Calculated in a Reference
Article)")
legend('x_2 at 50 kPa ( I_D )','x_2 at 101.325 kPa ( I_D )',['x_2 at 2' ...
    '00 kPa ( I_D )'],'x_2 at 50 kPa ( E_D )',['x_2 at 101.325 kPa ( E' ...
    '_D )'],'x_2 at 200 kPa ( E_D )')
```

[vi] MATLAB Code for Task 2C:

```
% TASK 2C

% clear

% Below, MATLAB's Import Data Function is Used to Transform the Data
% into a Manipulable Array:

% Set up the Import Options and import the data
opts = delimitedTextImportOptions("NumVariables", 6);
% Specify range and delimiter
opts.DataLines = [2, Inf];
opts.Delimiter = ",";
```

```
 % Specify column names and types
 opts.VariableNames = ["site_code", "measurement_id", "TIME", "TEMP",
"Press_ATM", "CO2EQ_PPM"];
 opts.VariableTypes = ["categorical", "double", "string", "double", "double",
"double"];
 % Specify file level properties
 opts.ExtraColumnsRule = "ignore";
 opts.EmptyLineRule = "read";
 % Specify variable properties
 opts = setvaropts(opts, "TIME", "WhitespaceRule", "preserve");
 opts = setvaropts(opts, ["site_code", "TIME"], "EmptyFieldRule", "auto");
 % Import the data
 Project_Data = readtable("/Users/samanthadufumulleruri/Documents/UCL/YEAR
1/ENGF0003 - Mathematical Modelling and Analysis/Assessments/2 MATLAB
COURSEWORK 2/DATA/Project_Data (1).csv", opts);
 % Clear temporary variables
 clear opts
 % Display results
 Project_Data

 % In the dataset CO2EQ_PPM (parts per million) is the value of y2 (the air
 % concentration of CO2) while Press_ATM is Pressure (Pa).
 % Additionally, all variables are expressed in their standard units.

 % Below, Data Points From the Table Above (Project_Data) are Converted
 % into Arrays:

 % The code below isolates only the data rom the Great Barrier Reef.
 project = Project_Data(Project_Data.site_code == "GBRWIS",:);

 % Fill Missing Data:
 project = fillmissing(project,"previous")

 % Convert from kPa (which are the units of pressure used in the column of
 % values: 'Press_ATM') to Pa (also convert temperature to Kelvin)).
 pressure = (project.Press_ATM).*10^3;
 temperature = (project.TEMP)+273.15;
 y_2 = project.CO2EQ_PPM;

 % Defining an Array for the Incompatible Datetime Values in the Raw Data:
 time0 = sortrows(project.TIME);
 time = datetime((time0), 'InputFormat','yyyy-MM-dd''T''HH:mm:ss''Z',...
     'TimeZone','UTC');

 % The function file (Task2_Function_Values) from task 2A (appendix iv) is
 % used to calculate H_21, phi_1 & phi_2 as below.
```

```
[H_21_2c, ~, ~, phi_2_2c] = Task2_Function_Values(pressure,temperature);

% Equation below is taken from equation 1.2 seen in the essay above (1)
% where x_2*H_21 = y_2*phi_2*pressure.
x_2_q2c = (y_2.*phi_2_2c.*pressure)./H_21_2c;

figure
% Cleaning Outliers (0.4 is a sensible replacement value for the outlier)
x_2_q2c(x_2_q2c>0.75) = 0.4
plot(time,x_2_q2c)
xlabel("Time (years)")
ylabel("Mass Fraction of CO_2 Dissolved in Water, x_2 (ppm?)")
title("Figure 9. Graph of x_2 Against Time")

% The 4x4 grid of graphs below are plots of y_2, phi_2, H_21 & pressure
% against time for a comprhensive analytical evaluation of relevent data
% trends.
hold off
figure
subplot(2,2,1)
% Cleaning Outliers (709 is a sensible replacement value for the outlier)
% y_2(y_2>1200) = 709
plot(time,y_2)
xlabel("Time (years)")
ylabel("Mass Fraction of CO_2 in Gas Form, y_2 (ppm)")
title("Figure 10. Graph of y_2 Against Time")

hold on
subplot(2,2,2)
% Cleaning Outliers (0.99 is a sensible replacement value for the outlier)
phi_2_2c(phi_2_2c>0.995) = 0.994472
plot(time,phi_2_2c)
xlabel("Time (years)")
ylabel("Fugacity in the Vapour Phase of C0_2, phi_2 (ND?)")
title("Figure ??. Graph of phi_2 Against Time")

subplot(2,2,3)
% Cleaning Outliers (160121000 is a sensible replacement value for the
% outlier)
H_21_2c(H_21_2c>2*10^8) = 160121000
plot(time,H_21_2c)
xlabel("Time (years)")
ylabel("Temperature Dependent Henry’s Constant, H_21 (ND?)")
title("Figure ??. Graph of H_2_1 Against Time")

subplot(2,2,4)
plot(time,pressure)
```

```
xlabel("Time (years)")
ylabel("Pressure (Pa)")
title("Figure ??. Graph of Pressure Against Time")
```

[vii] MATLAB Code for Task 3A

```
% Task 3A

% Use MATLAB to solve the system of ordinary
% differential equations shown in Eqs. 5.1 – 5.6
% using the reaction rates given in Table 3 and the
% initial conditions outlined in page 11. Plot your
% time-dependent solutions for all chemical species
% in a single figure.

clear

% The initial conditions when gaseous CO2 concentration ([CO2(g)]/[co2_g])is
% 0.065 M (Molarity or mol/l), temperature is 25 degrees celcius, time is 0s
% (seconds) and pressure is 1 atm (atmospheres) are as below:

co2_g = 0.065
co2_aq_0 = 5.41*10^-4
h2co3_0 = 1.64*10^-6
hco3_0 = 3.28*10^-4
co3_0 = 1.97*10^-8
h_0 = 1*10^-6

% Setting up the Variables Used in the ODE System Solver (ode15s):
% Tspan is 100 seconds as that's the time taken for the reaction to reach
% equilibrium.
tspan = [0 100]
% The initial conditions for the chemical reactants (R1-R4 in the brief) are
% as below.
y0 = [co2_g, co2_aq_0, h2co3_0, hco3_0, co3_0, h_0]

% In the ode15s function below dydt (ODEsys_reaction(t,y_con_array)) and y0
% are arrays of length 6 so that we can holistically integrate the system of
% 6 non-linear differential equations (equations 5.1 - 5.6 in the brief).
% Relative and Absolute tolerances are set as below to account for the
% small initial conditions (10^-7 and smaller).
% Additionally, The large variation of the reaction rates above means that a
% stiff solver like ode15s is suitable.
opts = odeset('RelTol',1e-6,'AbsTol',1e-12);
[t, y_con_array] = ode15s(@ODEsys_reaction,tspan,y0,opts);
```

```
% The use of the loglog function improves graph readablility.
figure
loglog(t,y_con_array, LineWidth=1.5)
legend('CO_2(g)', 'CO_2(aq)', 'H_2CO_3', 'HCO_3^-', 'CO_3^{2-}', 'H^+')
xlabel("Time (s)")
ylabel("Chemical Concentrations (mol/l)")
title("Graph of Concentrations of All Chemical Species by Time")



% The code below creates a system of 6 functions (dydt) that contain
% unknowns (y_con_array(1-6)) which can be input into the ode15s function'to
% find values of t & y_con_array numerically.

function dydt = ODEsys_reaction(t, y_con_array)

% Saving the forward reaction rate constants as column 1 in the arrays
% below and the reverse rate constants as column 2. The reverse rates for
% k3 & k4 have units M^-1s^-1 while the other constants are in s^-1.
k1_fr = [1*10^10 1*10^10]
k2_fr = [6*10^-2 2*10^1]
k3_fr = [1*10^7 5*10^10]
k4_fr = [3*10^0 5*10^10]

% Array y_con_array represents the values for the 6 chemical concentrations
% (in M) from equations 5.1 - 5.6 in the brief and are modelled as below:
dy1dt_co2g = -k1_fr(1)*y_con_array(1)+k1_fr(2)*y_con_array(2);
dy2dt_co2aq = k1_fr(1)*y_con_array(1)-(k1_fr(2)+k2_fr(1))*y_con_array(2)+...
    k2_fr(2)*y_con_array(3);
dy3dt_h2co3 = k2_fr(1)*y_con_array(2)-(k2_fr(2)+k3_fr(1))*y_con_array(3)+...
    k3_fr(2)*y_con_array(6)*y_con_array(4);
dy4dt_hco3 = k3_fr(1)*y_con_array(3)-
k3_fr(2)*y_con_array(6)*y_con_array(4)...
    -k4_fr(1)*y_con_array(4)+k4_fr(2)*y_con_array(6)*y_con_array(5);
dy5dt_co3 = k4_fr(1)*y_con_array(4)-k4_fr(2)*y_con_array(6)*y_con_array(5);
dy6dt_h = k3_fr(1)*y_con_array(3)-k3_fr(2)*y_con_array(6)*y_con_array(4)+
...
    k4_fr(1)*y_con_array(4)-k4_fr(2)*y_con_array(6)*y_con_array(5);

dydt = [dy1dt_co2g,dy2dt_co2aq,dy3dt_h2co3,dy4dt_hco3,dy5dt_co3,...
    dy6dt_h]';
end
```

[viii] MATLAB Code for Task 3B

```
% Task 3B

% Below is the code used to calculate the reaction rates of the compounds
% in 3A.
k1_fr = [1*10^10 1*10^10]
k2_fr = [6*10^-2 2*10^1]
k3_fr = [1*10^7 5*10^10]
k4_fr = [3*10^0 5*10^10]

% The reaction rates of the 6 chemical reactions (dydt):
dy1dt_co2g = -k1_fr(1).*y_con_array(:,1)+k1_fr(2).*y_con_array(:,2);
dy2dt_co2aq = k1_fr(1).*y_con_array(:,1)-(k1_fr(2)+k2_fr(1)).*y_con_array...
    (:,2)+k2_fr(2).*y_con_array(:,3);
dy3dt_h2co3 = k2_fr(1).*y_con_array(:,2)-(k2_fr(2)+k3_fr(1)).*y_con_array...
    (:,3)+k3_fr(2).*y_con_array(:,6).*y_con_array(:,4);
dy4dt_hco3 = k3_fr(1).*y_con_array(:,3)-k3_fr(2).*y_con_array(:,6).*...
    y_con_array(:,4)-
k4_fr(1).*y_con_array(:,4)+k4_fr(2).*y_con_array(:,6)...
    .*y_con_array(:,5);
dy5dt_co3 = k4_fr(1).*y_con_array(:,4)-k4_fr(2).*y_con_array(:,6).*...
    y_con_array(:,5);
dy6dt_h = k3_fr(1).*y_con_array(:,3)-k3_fr(2).*y_con_array(:,6).*...
    y_con_array(:,4)+k4_fr(1).*y_con_array(:,4)-
k4_fr(2).*y_con_array(:,6)...
    .*y_con_array(:,5);

% Plotting a log graph for comparison.
figure
subplot (1,2,1)
semilogx(t, dy3dt_h2co3,LineWidth=1)
hold on
semilogx(t, dy4dt_hco3,LineWidth=1)
semilogx(t, dy5dt_co3,LineWidth=1)
semilogx(t, dy6dt_h,LineWidth=1)
ylim([-1*10^(-4) 20*10^(-4)])
legend('H_2CO_3', 'HCO_3^-', 'CO_3^{2-}', 'H^+')
title("Reaction Rate Against Time")
xlabel("Time (s)")
ylabel("Reaction Rate (mol/dm)")

hold off
subplot (1,2,2)
semilogx(t, dy1dt_co2g,LineWidth=1)
hold on
semilogx(t, dy2dt_co2aq,LineWidth=1)
legend('CO_2(g)', 'CO_2(aq)')
```

```
title("Reaction Rate Against Time")
xlabel("Time (s)")
ylabel("Reaction Rate (mol/dm)")
```